MMML: Multimanifold Metric Learning for Few-Shot Remote-Sensing Image Scene Classification

Xiliang Chen, Guobin Zhu, and Jiaxin Wei

***Abstract*— Deep neural networks driven by large amounts of annotated samples have been successfully applied in remote-sensing scene classification (RSSC). On most common public datasets, the accuracies of scene recognition tasks have been close to saturation. A learning paradigm that specifically tackles this issue has emerged, that is, few-shot learning (FSL). FSL methods for natural image recognition are developing rapidly. Recently, these methods have been widely used in RSSC. Compared with natural images, the problems of intraclass differences and interclass similarities in RSSC are more serious, which hinders the further development of FSL methods for RSSC. To address this issue, we propose a multimanifold metric learning (MMML) framework for RSSC with FSL. Specifically, we use a lightweight convolutional neural network (CNN) as a feature extraction block. To further enhance the representation ability of features, we embed the feature map into two heterogeneous and complementary Riemannian manifold geometric structures, that is, the Grassmannian manifold and symmetric positive-definite (SPD) manifold. Then, to facilitate the measurement, we introduce the Riemann kernel function to embed two heterogeneous manifold structures into the high-dimensional Hilbert space for fusion. Finally, we design a learnable distance metric scheme that can be optimized according to the divergence of interclass pairs and intraclass pairs, thereby reducing the impact of large intraclass differences and high interclass similarity on scene recognition. We verify the effectiveness of MMML on three commonly used datasets. The results show that the accuracy of our proposed method is from 1.21% to 3.12% higher than the state-of-the-art methods.**

***Index Terms*— Multimanifold metric learning (MMML), remote-sensing scene classification (RSSC), Riemannian manifold.**

I. Introduction

REMOTE-SENSING scene classification (RSSC) is an important application developed from natural image recognition, which is the foundation of remote-sensing image interpretation. Different from pixel-by-pixel classification, the purpose of RSSC is to assign a preset scene category for the whole image according to the color, texture, spatial arrangement of geographical entities, and potential semantic information. RSSC has important application value in many industries, such as environmental pollution monitoring, disaster assessment, urban planning, and illegal building identification [1], [2], [3], [4], [5]. RSSC technology is improved with the development of deep-learning technology. Many excellent RSSC methods are constructed based on classical convolutional neural networks (CNNs), for example, AlexNet [6], [7], GoogleNet [8], ResNet [9], and DenseNet [10]. In recent years, the performance of RSSC algorithms has been significantly improved, and the classification accuracy (CA) on many public datasets has exceeded 90% [11]. However, these methods rely heavily on the quality and quantity of labeled samples. The more complex the network structure is, the larger the parameter scale is, and the more samples are required. As is known to all, annotated samples are a very scarce resource in remote sensing. On the one hand, labeling samples is a tedious job. In addition to consuming large amounts of time, the staff should be equipped with rich experience and professional knowledge. When there are unseen scene categories that require to be identified, we need to collect a large number of samples and train the model from scratch, which is unacceptable in some cases. For example, when natural disasters occur, we need to quickly identify damaged areas to propose a reasonable treatment plan. It is obvious that the image interpretation information obtained by this mode is lagging behind.

Humans have the ability to learn quickly and can master the features of a new category based on only one or several example pictures. Inspired by the human cognitive process of new things, researchers have proposed the concept of few-shot learning (FSL) [12]. The purpose of the FSL task is to train a classifier under only a few labeled samples so that it can recognize unseen samples with new categories. The FSL problem can be expressed in the form of N-way K-shot, where N represents the categories of the scene and K represents the number of samples in each category. Five-way one-shot and five-way five-shot are the two most representative forms, so they are widely studied in image recognition. According to different learning mechanisms, we classify the FSL methods into three categories: metric-based methods [13], [14], [15], [16], transfer-based methods [17], [18], [19], and optimization-based methods [20], [21], [22]. Among these methods, the metric-based method is the simplest and most effective, which has attracted the

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attention of many researchers [23]. Hence, most FSL methods for RSSC are based on metric learning [24], [25], [26].

The metric-based FSL methods have two key points: the representation of categories and the measurement of distance. Snell et al. [15] proposed the most classical metric-based FSL method. They use prototypes to represent the features of each category, define the average value of all features of each category in the embedding space as the prototype, and then predict the category according to the similarity between the prototype and the sample in the query set. However, in FSL, the number of samples in each category is only a few, and the mean of these features makes it difficult to represent the distribution of this category. Hence, researchers have proposed many methods to enhance the prototype. Zheng et al. [27] calculated the importance weight according to the difference between each sample and other samples and weighted each sample to obtain the final prototype. Ren et al. [28] introduced unlabeled samples to increase the number of samples and avoid the problem of poor representativeness of prototypes. In addition to using the features of the support set to calculate the prototype, Cheng et al. [24] also regarded the confidence score of the query sample as another prototype and used the interactive information between the support set and query set to further calibrate the prototype, which performed well in the RSSC task. However, in these methods, the prototype is a first-order statistical feature calculated on the Euclidean space composed of vectors, and the representation ability is limited. Li et al. [29] used the covariance matrix to represent the prototype. The covariance matrix is a second-order statistical information with manifold structures, which can capture the low-level distribution information of each category and has good robustness. Wang et al. [30] thought that the Riemannian manifold has excellent modeling capabilities for video frame sequences, and the performance exceeds the linear modeling on Euclidean space. Inspired by these works, we propose a prototype representation method based on the multimanifold space. The feature map is embedded into two heterogeneous and complementary Riemannian manifold spaces, that is, the Grassmannian manifold and symmetric positive-definite (SPD) manifold, and the prototype on the manifold space has been defined. To facilitate the measurement, we introduce the Riemann kernel function to embed two heterogeneous manifold structures into the high-dimensional Hilbert space for fusion. Another key point of the metric-based method is the distance metric function. Traditional methods [13], [15] use feature-independent distance metric methods, such as Euclidean distance and cosine distance. These methods assume that the features used to calculate the distance are irrelevant and have uniform variance. However, it is difficult to ensure that the distribution of features conforms to this assumption in practical applications. Hence, Bateni et al. [31] used the Mahalanobis distance that can represent the correlation between vectors as the distance function. The vectors here represent the global features of the image. To further capture the local features of the image, Zhang et al. [32] expressed the features as tensors and used the Earth mover's distances which can reflect the correlation between tensors to calculate the similarity. Furthermore, Wang et al. [33] proposed a method to

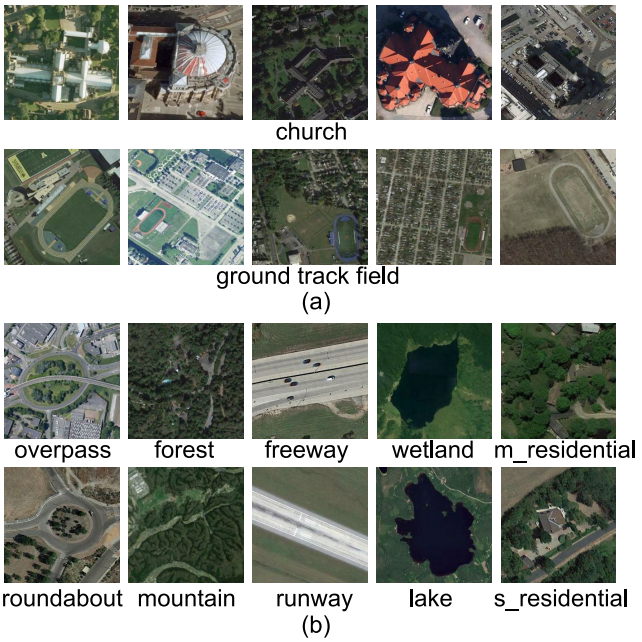


Fig. 1. (a) Intraclass image pairs that have great differences. (b) Interclass image pairs that have great similarities. All image pairs are obtained from the NWPU dataset.

adaptively learn the similarity between features, that is, using a neural network as a distance function. These methods have achieved good results in natural image recognition. Compared with natural images, the problems of intraclass difference and interclass similarity in RSSC are more serious. As shown in Fig. 1, both the church and the ground track field in Fig. 1(a) have strong intraclass differences. The same category of the scene shows great differences in color, texture, and spatial arrangement of geographical entities due to the influence of illumination, shooting angle, and scale. The first and second lines in Fig. 1(b) are image pairs with different categories, which are similar in appearance. These two phenomena seriously affect the accuracy of scene recognition. Hence, the calculation method of distance is very important. To tackle this issue, we design a learnable distance metric scheme, which can optimize the distance according to the divergence of interclass pairs and intraclass pairs, to reduce the influence of intraclass variance and interclass variance on scene recognition.

Considering the above factors, we propose a multimanifold metric learning (MMML) framework for RSSC with FSL. Specifically, we use a lightweight CNN with four layers as a feature extraction block. To further enhance the representation ability of features, we embed the feature map into two heterogeneous and complementary Riemannian manifold geometric structures, that is, the Grassmannian manifold and SPD manifold. Then, to facilitate the measurement, we introduce the Riemann kernel function to embed two heterogeneous manifold structures into the high-dimensional Hilbert space for fusion. Finally, we design a learnable distance metric scheme that can be optimized according to the divergence of interclass pairs and intraclass pairs. We use the embedding features of the samples in the query set and the embedding features of the same category in the support set to construct intraclass

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pairs and calculate the intraclass divergence. In addition, the embedding features of the samples in the query set and other embedding features that do not belong to this category in the support set are used to construct interclass pairs to calculate interclass divergence. The ratio of interclass divergence and intraclass divergence is used as an optimization function for calculating distance, we should maximize it, thereby reducing the impact of large intraclass differences and high interclass similarity on scene recognition. Our contributions can be summarized as follows.

1) We propose a few-shot RSSC framework that combines the CNN and MMML. In the framework, two heterogeneous and complementary Riemannian manifold features based on CNN local features are fused to enhance the representation ability. Two prototypes are defined on the different manifold space as the representations of the categories and we explore the optimal weight factors for integrating them.
2) We design a learnable distance metric scheme that can be optimized according to the divergence of interclass pairs and intraclass pairs. The ratio of interclass divergence and intraclass divergence is used as an optimization function for calculating distance, we should maximize it, thereby reducing the impact of large intraclass differences and high interclass similarity on scene recognition.
3) We conduct comparative experiments on three public datasets to investigate the performance of our proposed method. The results demonstrate that MMML is superior to the previous methods in the task of few-shot RSSC.

The rest of this article is organized as follows: In Section II, we will introduce the work related to RSSC, RSSC with FSL, and Riemannian manifold metric learning. In Section III, we will introduce the detailed methodology of the MMML framework. Experimental results on three public datasets are presented in Section IV. Finally, in Section V, we will draw a conclusion.

II. Related Work

A. Remote-Sensing Scene Classification

Deep-learning technology is known for its powerful feature extraction capabilities and has become a research hotspot in recent years, which has been widely applied to RSSC [11]. According to the usage of labeled and unlabeled samples in the training process, the model of RSSC can be summarized as supervised learning, unsupervised learning, semisupervised learning, and self-supervised learning. Among them, the combination of supervised learning and CNN is the most common way [33], [34], [35], [36], [37]. However, these methods usually require a large number of labeled training samples. In remote sensing, the cost of labeling samples is very high, which makes it difficult to obtain training samples. Unsupervised learning can extract features without label information. Based on the nearest neighbor principle, features with high similarity are judged as the same category [38], [39], [40]. However, this method lacks the constraints of tags, resulting in a high recognition error rate. Semisupervised learning can utilize both labeled and unlabeled data, which can reduce the dependence on labeled samples [41], [42]. Self-supervised learning is a new machine-learning paradigm that has emerged in recent years. The core difference between self-supervised learning and supervised learning methods is that self-supervised learning mines its own supervision information from massive unlabeled data through artificially designed self-supervised learning signals to form pseudo-labels, thereby replacing traditional manually labeled data to drive the model for feature learning [43]. Some researchers have applied the self-supervised learning model to RSSC [44], [45], [46]. In addition, considering that natural images have rich labeled samples and the feature extraction process of natural images and remote-sensing images is similar, the features and parameters learned from natural images can be adapted in remote sensing. Based on this principle, researchers have proposed transfer learning. There are also many successful applications of transfer learning in RSSC [47], [48]. Although these methods have been considered to reduce the dependence on labeled samples, they still cannot be trained under a few labeled samples. In addition, these methods are trained on the dataset with preset categories. When encountering unseen categories, it does not work. Our proposed MMML has this capability, which is the fundamental difference between our method and the traditional RSSC methods.

B. FSL for RSSC

In recent years, FSL has become a research hotspot in RSSC. Due to the complexity of remote-sensing images, the model developed for the natural image recognition task has limited performance on RSSC. Therefore, researchers have proposed a series of excellent methods for RSSC. At present, there are mainly two kinds of RSSC methods based on FSL, that is, optimization-based methods and metric-based methods. Because the former needs to be fine-tuned on the test sample, it has rarely been applied. Alajaji and Alhichri [49] optimized the hyperparameters in MAML [22] to improve its performance in RSSC tasks. Ma et al. [50] used the unsupervised domain adaptation strategy to realize knowledge transfer, then the RSSC task with FSL can be implemented on a deeper network structure. Zhai et al. [51] proposed a lifelong learning method based on meta-learning. Metric-based methods are simple and effective, so they are more widely used in remote sensing. Cheng et al. [24] proposed SPNet, in addition to using the features of the support set to calculate the prototype, the confidence score of the query sample is also regarded as another kind of prototype, and the information interaction between the support and the query sample is used to further calibrate the prototype, which has achieved high performance in RSSC. Li et al. [26] used channel attention and spatial attention modules to automatically discover the distribution areas of features that contribute more significantly to scene classification. In addition, they also combined the features of the support set and the query set to adaptively select image pairs with stronger semantic relevance to calculate the similarity, to promote RSSC in the case of limited training samples. Li et al. [52] proposed an FSL method based on an attention metric network. In addition to using spatial attention

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and channel attention modules to fuse different features, cross-attention modules can also be used to highlight different kinds of features, which has greatly improved the performance of the similarity metric. Li et al. [53] maximized the distance between different categories of samples by constraining the loss function, which can provide clearer segmentation boundaries for distinguishing different types of features. A learnable distance metric is also proposed to make it more adaptive. Cui et al. [54] designed a meta-kernel network to map indistinguishable features in low-dimensional space to high-dimensional space through kernel functions so that the boundaries of each category are more obvious. At the same time, they designed a new loss function that can minimize the intraclass and interclass variance ratio, thereby reducing the misclassification caused by large intraclass variance and small interclass variance. The proposed MMML is a metric-based method. Different from the above method measured on Euclidean space, we map features to two heterogeneous and complementary Riemannian manifold geometric structures to enhance their representation ability and then embed them into high-dimensional Hilbert space for fusion by Riemannian kernel function. In addition, we design a scheme that can optimize the distance according to the divergence of interclass pairs and intraclass pairs, to reduce the influence of intraclass variance and interclass variance on scene recognition.

C. Riemannian Manifold Metric Learning

Riemannian manifold is a special differential manifold, that is, a differential manifold with Riemannian metric. Riemannian manifold can be divided into many kinds. The most classical Riemannian manifolds are the SPD manifold and the Grassmann manifold. Riemannian metric is the most important property of the Riemannian manifold. The geodesic distance derived from the Riemannian metric is the most essential geometric metric to measure the distance between two points on the Riemannian manifold, which has important applications in image processing [55]. Masci et al. [56] proposed a geodesic CNN based on Riemannian manifold space, which achieves the most advanced performance in the fields of shape description and retrieval. Wang et al. [57] constructed an SPD manifold directly based on the covariance matrix descriptor of the image set and then designed an end-to-end SPD manifold neural network and successfully used it for image set classification. Inspired by the kernel function in Euclidean space, Jayasumana et al. [58] developed a kernel function for the SPD manifold and proved that the manifold kernel method is superior to the corresponding Euclidean kernel method. Wang and Wu [59] extended the idea of deep learning in Euclidean space to Grassmann manifold space. They designed a simple Grassmann manifold feature learning network (GrasNet), which provides a new method for image set classification. Some researchers have also constructed an image recognition framework with FSL based on the Riemannian manifold. For example, Li et al. [29] proposed a covariance metric network. They used the feature maps of the CNN to construct covariance descriptors, which are essentially SPD manifolds. Then, the consistency of distribution between query samples and each category is further measured based on the SPD manifold. Inspired by the above work, we propose a multimanifold learning framework for RSSC with FSL. Compared with [29], our innovations are in three aspects. First, we simultaneously embed features into two heterogeneous and complementary Riemannian manifolds to enhance the representation ability of features. Second, to fuse two different Riemannian manifold structures, we use the Riemannian kernel function strategy to map them into a unified high-dimensional space for further processing. Finally, considering that remote-sensing images have complex intraclass differences and interclass similarities, we optimize the distance by maximizing the ratio of interclass divergence to intraclass divergence, thereby reducing the impact of this phenomenon on scene recognition.

III. Proposed Method

A. Definition of Few-Shot Problem

Currently, the few-shot problem is usually trained and tested based on the cross-task mechanism under meta-learning, in which each task is called a meta-task and follows the standard N-way K-shot paradigm, where N represents the number of classes and K represents the number of labeled samples per class. In other words, the model learns from only K labeled samples per class in a meta-task, and after training, it can recognize the remaining data from N classes.

In few-shot image recognition, a meta-task $\mathfrak{T}$ refers to an N-way K-shot task scenario, where the task scenario is sampled from a specific dataset $\mathfrak{D}$, and a meta-task's dataset consists of a support set ($\mathfrak{S}$) and a query set ($\mathfrak{Q}$). $\mathfrak{S} = \{(x_i, y_i)\}_{i=1}^{N\times K}$, $\mathfrak{Q} = \{(x_i, y_i)\}_{i=1}^{N\times q}$, $\mathfrak{T} = \{(\mathfrak{S}_i, \mathfrak{Q}_i)\}_{i=1}^{m} \in \mathfrak{D}$, where x_i represents an image, y_i represents its corresponding label, N represents the number of image categories, K represents the number of images of each category in the support set, q represents the number of images of each category in the query set, and m represents the number of meta-learning tasks. The entire sampling process involves randomly sampling N categories from the dataset $\mathfrak{D}$, then randomly selecting K samples from each category as the support set, and finally randomly selecting q samples from the remaining samples in each category as the query set.

B. Descriptors on Grassmannian Manifold

A manifold is a smooth surface embedded in a high-dimensional Euclidean space. All m-dimensional linear subspaces $R^m (0 \le m \le d)$ of a d-dimensional space R^d constitute a Grassmann manifold, denoted as $G(m, d)$ [60]. It is a special Riemannian manifold, generally represented by the orthogonal basis of the subspace. According to [61], we can express the Grassmannian manifold as follows:

$$G(m,d) = \left\{ G \in R^{d\times m} : G^T G = I_m, GG^T = M \right\} \quad (1)$$

where I_m is an identity matrix with the size of $m \times m$ and M is a shared matrix for a certain group of G with the size of $d \times d$. The Grassmannian manifold $G(m, d)$ is a quotient space, each point of which is a linear subspace formed by a set of matrices with orthonormal columns [63]. The columns

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of each matrix of an equivalent class span the same linear subspace. In the framework of the projection mapping $\boldsymbol{\phi}(G) = GG^T$, GG^T is used to represent the basic elements on the Grassmann manifold, that is, the linear subspace span(G). It is a one-to-one, continuous, and differentiable mapping, and the inverse mapping is also continuous and differentiable. Therefore, each element on the Grassmannian manifold corresponds to only one projection matrix.

Given a vector sequence $\{v_1, v_2, \ldots, v_n\}$, where $v_n \in R^d$ denotes a column vector with d-dimensional. They can form a matrix $V \in R^{d\times n}$. Furthermore, it can be described by the m-dimensional linear subspace span(U) of the orthogonal basis matrix $U \in R^{d\times m}(0 \le m \le d)$. The linear subspace of matrix V can be obtained by the singular value decomposition (SVD)

$$VV^T \simeq U\Lambda U^T \tag{2}$$

where the diagonal values of Λ represent the m largest eigenvalues, and the orthogonal matrix U represents the corresponding eigenvectors, respectively, where $U^TU = I_m$, and I_m is an identity matrix with the size of $m \times m$.

The Grassmann manifold is a nonlinear space, and the distance metric method in Euclidean space cannot be directly applied to the Grassmann manifold. By studying the specific Riemannian geometric characteristics of the Grassmann manifold, the projection mapping framework is introduced to represent the basic elements of the Grassmann manifold as the corresponding projection matrix [62]. The projection mapping function $\boldsymbol{\phi}(G)$ is a symmetric matrix with the dimension of $d \times d$, and the corresponding inner product can be defined as $< G_1, G_2 >_{\boldsymbol{\phi}} = \mathrm{tr}(\boldsymbol{\phi}(G_1)^T, \boldsymbol{\phi}(G_2))$. This inner product is invariant for linear subspaces, and the distance can be derived as

$$D_G(G_1, G_2) = \frac{1}{\sqrt{2}}\left\|G_1G_1^T - G_2G_2^T\right\|_F \tag{3}$$

where $\|.\|_F$ denotes the Frobenius norm of a matrix. Then, the Grassmann kernel can be defined as follows according to [62]:

$$K_G\left(G_1G_1^T, G_2G_2^T\right) = \mathrm{tr}\left[\left(G_1G_1^T\right)\left(G_2G_2^T\right)\right] = \left\|G_1^TG_2\right\|_F^2. \tag{4}$$

C. Descriptors on the SPD Manifold

A real $d \times d$ matrix C is SPD if and only if $z^TCz > 0$ for every nonzero vector $z \in R^d$. The space of real $d \times d$ SPD matrices, represented by S_{++}^d forms a Lie group manifold structure [63]. An SPD matrix is equivalent to a point on the SPD manifold. Suppose $S_1, S_2 \in S_{++}^d$, to evaluate the positional relationship between S_1 and S_2 on the manifold, then a metric method is needed to calculate the geodesic distance of S_1 and S_2. There are many excellent works devoted to solving the problem of distance metrics on SPD manifold space, such as the affine invariant Riemannian metric (AIRM) [64], the log-Euclidean Riemannian metric (LERM) [64], the Jensen–Bregman LogDet (JBLD) [65], and the symmetrized KL-divergence metric (KLDM) [66]. Considering that LERM is not only easy to calculate, but also maintains many characteristics of affine invariant metric. So, we use it to measure the SPD manifold. It can be expressed as follows:

$$D_S(S_1, S_2) = \|\log(S_1) - \log(S_2)\|_F \tag{5}$$

where $\|.\|_F$ denotes the Frobenius norm of a matrix. Wang et al. [67] proposed a Riemannian kernel on the SPD manifold, which can be expressed as follows:

$$K_S(S_1, S_2) = \mathrm{tr}\left[\log(S_1)\log(S_2)\right] \tag{6}$$

where tr[.] denotes the trace of the matrix.

Region covariance descriptors are SPD matrices and therefore can be analyzed through Riemannian geometry. Hence, we embed the features into the SPD manifold space by constructing covariance matrices. Given a vector sequence $\{v_1, v_2, \ldots, v_n\}$, where $v_n \in R^d$ denotes a column vector with d-dimensional, the covariance matrix C can be defined as

$$C = \frac{1}{n-1}\sum_{i=1}^{n}(v_i - \mu)(v_i - \mu)^T \tag{7}$$

where μ is the mean of n samples, $\mu \in R^d$, and $C \in R^{d\times d}$.

D. MMML With Kernel Function Strategy

In Fig. 2, after the image is input into the embedding model, we finally get a feature map with the dimension of $C \times H \times W$. First, we construct the SPD manifold through the region covariance descriptor. We regard each pixel as a local feature, the pixels on the same position of different channels are concatenated together to form a $1 \times C$ vector, the total number of vectors is $H \times W$, then calculate the covariance matrix between them, and the covariance descriptor is obtained, which dimension is $H \times W$. Then, we construct the Grassmannian manifold. We flatten each feature map to form a $1 \times HW$ vector and recombine these vectors into a matrix with the dimension of $HW \times C$. According to (2), we can map the matrix on the Grassmann manifold space. In the metric model, we use the covariance matrix of the mean of each category of the feature map as the prototype. Then the prototype is mapped to the SPD manifold and the Grassmann manifold, which are represented by P_S and P_G, respectively, we call them the prototypes on manifold space. We construct intraclass pairs between the Riemannian manifold descriptors in the query set and the same category of descriptors corresponding to the support set and construct interclass pairs between the Riemannian manifold descriptors in the query set and other different categories of descriptors in the support set. Our goal is to use interclass divergence and intraclass divergence to optimize the distance. We use $\mathcal{T}^l = [\mathcal{T}_1^l, \mathcal{T}_2^l, \ldots, \mathcal{T}_N^l]$ to represent the features of each group of positive and negative samples on Riemannian manifold space. $\mathcal{T}_i^l \in R^{d_l}$ denotes the feature descriptor of the lth Riemannian manifold. N denotes the total number of positive and negative samples, and l denotes the type of Riemannian manifold, which ranges from 1 to L. We set $L = 2$. In addition, since the two types of Riemann descriptors are heterogeneous, they cannot be directly fused. Inspired by Wang et al. [30] and Chen et al. [68], we use the kernel

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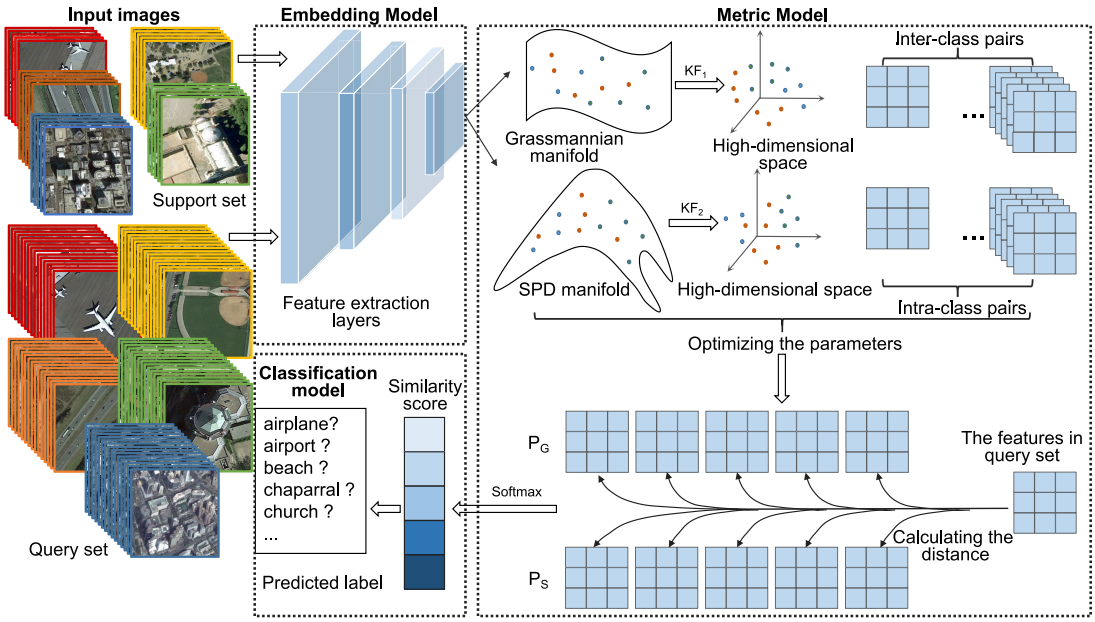


Fig. 2. Overall framework of MMML. P_S and P_G represent prototypes on the SPD manifold and Grassmann manifold, respectively.

function to embed them into Hilbert space and then use dot product operation to simplify the calculation process. We use $\mathbf{\Phi}_i^l$ to denote the features on Hilbert space which is generated from $\mathcal{T}_i^l$. This mapping function can be expressed as $\mathbf{\Phi} : R^{d_l} \longrightarrow \mathcal{H}$, where R^{d_l} represents the lth Riemannian manifold space and $\mathcal{H}$ the Hilbert space. Although the specific expression of $\mathbf{\Phi}$ is implicit, to describe the calculation process, we regard it as a known function and finally use the dot product of the kernel function to calculate. In summary, for any given prototype P_i in a support set and an image E_j in a query set, the distance metric on $\mathcal{H}$ can be defined as

$$d(P_i, E_j) = \mathrm{tr}\left[\sum_{l=1}^{L} \lambda_l \left(\mathbf{\Phi}_i^l - \mathbf{\Phi}_j^l\right)^T D_M \left(\mathbf{\Phi}_i^l - \mathbf{\Phi}_j^l\right) \lambda_l\right] \tag{8}$$

where λ_l denotes the weight factor of the lth manifold, which is a hyperparameter and needs to be determined by repeated experiments. D_M is a Mahalanobis matrix that needs to be optimized. Since it is a symmetric positive semidefinite matrix, we can reconstruct it through matrix decomposition, that is, $D_M = WW^T$, and W is a nonsquare matrix. Hence, (8) can be rewritten as follows:

$$d(P_i, E_j) = \mathrm{tr}\left[W^T \left(\sum_{l=1}^{L} \lambda_l \left(\mathbf{\Phi}_i^l - \mathbf{\Phi}_j^l\right)\left(\mathbf{\Phi}_i^l - \mathbf{\Phi}_j^l\right)^T \lambda_l\right) W\right]. \tag{9}$$

Since our purpose is to minimize the distance between intraclass sample pairs and maximize the distance between interclass sample pairs, we can define the following discriminant function for the nonfull rank matrix W:

$$W^* = \arg\max_W \mathcal{J}(W) = \arg\max_W \frac{O_b(W)}{O_w(W)} \tag{10}$$

where $O_b(W)$ and $O_w(W)$, respectively, denote the between-class dispersion and within-class compactness, which can be formulated as

$$\begin{cases} O_b(W) = \dfrac{1}{N_b} \displaystyle\sum_{i=1}^{N} \sum_{j:l_i \neq l_j} d(P_i, E_j) = \mathrm{tr}\left(W^T \Omega_b W\right) \\ O_w(W) = \dfrac{1}{N_w} \displaystyle\sum_{i=1}^{N} \sum_{j:l_i = l_j} d(P_i, E_j) = \mathrm{tr}\left(W^T \Omega_w W\right) \end{cases} \tag{11}$$

where N_b and N_w denote the number of negative and positive sample pairs, respectively, and l_i and l_j denote the category labels. Ω_b and Ω_w, respectively, denote the between-class and within-class scatter matrices, and their specific forms are as follows:

$$\begin{cases} \Omega_b = \dfrac{1}{N_b} \displaystyle\sum_{i=1}^{N} \sum_{j:l_i \neq l_j} \sum_{l=1}^{L} \lambda_l \left(\mathbf{\Phi}_i^l - \mathbf{\Phi}_j^l\right)\left(\mathbf{\Phi}_i^l - \mathbf{\Phi}_j^l\right)^T \lambda_l \\ \Omega_w = \dfrac{1}{N_w} \displaystyle\sum_{i=1}^{N} \sum_{j:l_i = l_j} \sum_{l=1}^{L} \lambda_l \left(\mathbf{\Phi}_i^l - \mathbf{\Phi}_j^l\right)\left(\mathbf{\Phi}_i^l - \mathbf{\Phi}_j^l\right)^T \lambda_l. \end{cases} \tag{12}$$

In fact, the calculation of (12) is difficult to be performed, because the form of $\mathbf{\Phi}$ is unknown. However, when we express the basis w_h as a linear combination of all the training samples in the feature space $\mathcal{H}$, that is,

$$w_h = \sum_{i=1}^{N} e_i^h \mathbf{\Phi}_i^l \tag{13}$$

where e_i^h are the coefficients, we can calculate through the kernel trick

$$\sum_{l=1}^{L} w_h^T \mathbf{\Phi}_i^l = \sum_{i=1}^{N} \sum_{l=1}^{L} e_i^h \left(\mathbf{\Phi}_i^l\right)^T \mathbf{\Phi}_i^l = \sum_{l=1}^{L} \left(e^h\right)^T K_i^l \tag{14}$$

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where $e^h \in R^{N\times 1}$ is a column vector and e_i^h is the ith entry of it. $K^l \in N \times N$ is generated by the lth Riemannian kernel function, and K_i^l denotes the ith column of it. Hence, we can transform the optimization of matrix W into the optimization of matrix E, then (10) can be rewritten as

$$E^* = \arg\max_E \mathcal{J}(E) = \arg\max_E \frac{\mathrm{tr}\left(E^T \Omega_b^* E\right)}{\mathrm{tr}\left(E^T \Omega_w^* E\right)} \tag{15}$$

where O_b^* and O_w^* can be expressed as

$$\begin{cases} \Omega_b^* = \dfrac{1}{N_b} \displaystyle\sum_{i=1}^{N} \sum_{j:l_i \neq l_j} \sum_{l=1}^{L} \lambda_l \left(K_i^l - K_j^l\right)\left(K_i^l - K_j^l\right)^T \lambda_l \\ \Omega_w^* = \dfrac{1}{N_w} \displaystyle\sum_{i=1}^{N} \sum_{j:l_i = l_j} \sum_{l=1}^{L} \lambda_l \left(K_i^l - K_j^l\right)\left(K_i^l - K_j^l\right)^T \lambda_l. \end{cases} \tag{16}$$

According to [69], the optimization problem of matrix E can be transformed into a relatively simple problem of calculating the matrix trace

$$\begin{aligned} E^* &= \arg\max_E \mathrm{tr}\left[\left(E^T \Omega_w^* E\right)^{-1}\left(E^T \Omega_b^* E\right)\right] \\ &= \arg\max_E \frac{\left|E^T \Omega_b^* E\right|}{\left|E^T \Omega_w^* E\right)|}. \end{aligned} \tag{17}$$

Then, matrix decomposition is performed on $(\Omega_w^*)^{-1}(\Omega_b^*)$ to obtain its eigenvalues and eigenvectors. The eigenvectors are arranged by the descending order of the corresponding eigenvalues, and the eigenvectors are regarded as column vectors to form a matrix, which is the coefficient matrix E.

E. Method Overview

Fig. 2 shows the overall framework of MMML, which consists of an embedding model, a metric model, and a classification model. The embedding model is composed of four CNN blocks, and each block contains a 3×3 convolution (Conv), a batch normalization (BN), a rectified linear unit (ReLU) activation nonlinearity layer, and a 2×2 max-pool layer. The convolutional kernel size is set to 3×3, the max pooling kernel size is set to 2×2, and the convolutional stride is set to 1. The metric model mainly includes the construction of multiple manifold descriptors and the metric of distance. First, we need to input the images from the query set and the support set into the embedded model for feature extraction and map the feature map of the last layer of the CNN to the Glassman manifold and the SPD manifold spaces according to the methods introduced in Sections III-A and III-B. Subsequently, we calculate the average value of feature maps in the support set with the same categories and map them to the Glassman manifold and SPD manifold spaces, respectively. We regard them as two prototypes, namely P_G and P_S. Finally, we construct positive sample pairs and negative sample pairs to optimize the distance between the samples in the query set and the prototypes. Specifically, we regard the sample pairs composed of the prototype and other categories of samples in the support set as interclass sample pairs, that is, negative sample pairs. We regard the sample pairs composed of the sample in the query set and the same category of samples corresponding to the support set as within-class sample pairs, that is, positive sample pairs. The parameters in the distance metric function are optimized according to the principle of maximizing interclass distance and minimizing intraclass distance. The classification model is mainly used in the test stage. For a given image with an unknown category, we calculate the distance between it and each prototype and predict its category according to the nearest neighbor principle.

IV. Experiments

A. Dataset Descriptions

In this article, to demonstrate the performance of our proposed method MMML, we performed the experiments on three public benchmark datasets, that is, the NWPU-RESISC45 (NWPU) dataset [70], the WHU-RS19 (WHU) dataset [71], and the UC Merced Land Use (UCM) dataset [72]. We split each dataset into the training set, validation set, and test set. To make a fair comparison with other methods, we follow the splitting strategy in [26] and [73]. Splitting results are shown in Table I. In addition, we adjust the size of all images to 224×224.

The NWPU dataset was published by Northwestern Polytechnical University in 2017. This dataset includes a total of 31 500 images with the size of 256×256, covering 45 scene categories, with 700 images in each category. The spatial resolution of the dataset is about 0.2–30 m. The dataset comes from many countries and regions, which is relatively a large-scale dataset at present. In addition, the images have great differences in translation, rotation, viewpoint, posture, lighting, background, and occlusion, which exist great intraclass differences and interclass similarities. To keep consistent with the previous FSL works for RSSC, we assigned samples to the training set, validation set, and test set at a ratio of 25:10:10.

The WHU is a group of satellite images collected from Google Earth released by Wuhan University, which can provide high-resolution satellite images up to 0.5 m. It can be used for scene classification and retrieval. The size of the image is 600×600. The dataset includes 19 categories of scenes, 50 images in each category, and a total of 1005 images. It is worth noting that the samples in the same category may be collected from different satellites and have different resolutions, so they may have different scales, directions, and illumination. For the WHU, nine, five, and five categories are divided into training, validation, and test sets.

The UCM is a classic RSSC dataset released by the Computer Vision Laboratory at the University of California, Merced, in 2010. The images are manually extracted from the National Map Urban Area Imagery of the National Geological Survey, which can be used to identify the category of the scene in urban areas. The spatial resolution of the dataset is about 0.3 m, and the image size is 256×256. The dataset includes 21 categories of scenes, 100 images in each category, and a total of 2100 images. For this dataset, 25 categories are regarded as a training set, ten categories are regarded as a

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TABLE I
SPLITS FOR THE TRAINING SET, VALIDATION SET, AND TESTING SET

Name	Training Categories	Validation Categories	Testing Categories
NWPU dataset	Wetland, Stadium, Sparse residential, Ship, Sea ice, Roundabout, Rectangular farmland, Railway, Palace, Mountain, Mobile home park, Meadow, Island, Lake, Harbor, Golf course, Freeway, Desert, Cloud, Church, Chaparral, Bridge, Beach, Baseball diamond, Airplane	Thermal power station, Terrace, Tennis court, Storage tank, Snowberg, Runway, Railway station, Overpass, Industrial area, Commercial area	River, Parking lot, Medium residential, Intersection, Ground track field, Forest, Dense residential, Circular farmland, Basketball court, Airport
WHU dataset	Residential, Port, Parking, Mountain, Industrial, Football field, Desert, Bridge, Airport	Railway station, Park, Forest, Farmland, Beach	Viaduct, River, Pond, Meadow, Commercial
UCM dataset	Parking lot, Overpass, Medium residential, Harbor, Freeway, Dense residential, Chaparral, Buildings, Baseball diamond, Agricultural	Storage tanks, Runway, Intersection, Forest, Airplane	Tennis court, River, sparse residential, Mobile home park, Golf course, Beach

validation set, and the other ten categories are regarded as a test set.

B. Implementation Details and Hyperparameter Settings

In this experiment, like other FSL methods, we adopt a meta-learning strategy for training. The minimum training unit of this pattern is episode. An episode is composed of a support set and a query set. A support set contains a dataset in the form of N-way K-shot. We perform the experiments in the case of five-way one-shot and five-way five-shot. The categories of the samples in the query set are the same as the support set, but there is no limit to the quantity of the samples in the query set. We set it to 15. To make a fair comparison with other methods, we follow [73] and [74] to set the number of episodes. To prepare the training dataset, we randomly generated 200 episodes for the training set, 500 episodes for the validation set, and 2000 episodes for the testing set. We use the average CA or CAs on the testing set with 95% confidence to evaluate our model. In addition, we also display the CA of each category through a confusion matrix on all test samples. We used the Adam optimizer to optimize the model, the initial learning rate is set to $1e^{-3}$, which is updated every ten epochs. In addition, we have three hyperparameters that need to be determined through several repeated experiments, that is, m (where m denotes the dimension of the linear subspace) in Section III-A and λ_l (where l denotes the type of Riemannian manifold) in Section II-C. Let λ_1 denotes the SPD manifold and λ_2 denotes the Grassmann manifold, then $\lambda_1 + \lambda_2 = 1$. Finally, for the NWPU, WHU, and UCM datasets, the values of m are set to 120, 100, and 120, respectively. In addition, λ_1 is set to 0.6 and λ_2 is set to 0.4. We will introduce the hyperparameter selection procedure in Section IV-E. We implement our model through the PyTorch framework. All the experiments are performed on a computer equipped with the NVIDIA RTX 3090 GPU.

C. Experimental Result Analysis

In this section, we verify the effectiveness of MMML on three public datasets (i.e., NWPU, WHU, and UCM).

TABLE II
CAS (%) OF MMML ON THE NWPU DATASET

Method	Five-way	
	One-shot	Five-shot
MN [13]	42.81±0.12	47.71±0.26
S2M2 [77]	63.24±0.47	83.23±0.28
Meta-SGD [75]	60.56±0.28	74.33±0.65
TPN [78]	66.51±0.87	78.50± 0.56
DeepEMD [32]	64.39±0.84	78.01± 0.56
RN [16]	66.05±0.32	78.24±0.47
IDLN [79]	**75.25± 0.75**	84.67 ± 0.23
PN [15]	41.57±0.42	63.09±0.08
DLA-MatchNet [26]	68.80±0.70	81.63±0.46
GES-Net [80]	70.83± 0.85	82.27± 0.55
MAML [20]	47.32±0.10	63.03±0.55
MKN [54]	65.84 ±0.89	82.67± 0.55
Baseline (Conv-4) [81]	62.21±0.48	79.13±0.32
SCL-MLNet [74]	62.21±1.12	80.86±0.76
LLSR [51]	51.43	72.90
TAE-Net [82]	69.13± 0.83	82.37±0.52
RS-MetaNet [53]	46.21±0.58	68.75±0.70
CovaMNet [29]	43.62±0.67	67.23±0.15
SIB [83]	67.34±0.81	78.28±0.48
TSC [76]	73.26±0.15	84.62±0.35
CAN+T [84]	69.89±0.58	81.04±0.33
SPNet [24]	67.84±0.87	83.94 ±0.50
PCFGNet [85]	72.05 ±0.75	**85.07 ± 0.45**
MMML (ours)	**77.35±0.73**	**88.19±0.47**

In addition, to demonstrate the performance of our proposed method, we compare the experimental results with the current mainstream methods. The experimental results are shown in Tables II–IV. Among these methods, only meta-SGD [75], MAML [20], and LLSR [51] belong to the optimization-based method, and the others belong to the metric-based method, which not only include the classical methods (e.g., MN [13], RN [16], pn [15], etc.) in computer vision, but also include the methods (e.g., DLA-MatchNet [26], MKN [54], SCL-MLNet [74], TSC [76], SPNet [24], etc.) developed in recent years for RSSC. To emphasize the comparison results more intuitively, we present the best and second best results with bold text.

Table II shows the CAs on the NWPU dataset. It is the most challenging dataset, which has the largest number of samples

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TABLE III
CAs (%) OF MMML ON THE WHU DATASET

Method	Five-way	
	One-shot	Five-shot
Meta-SGD [75]	51.78±1.05	65.47±0.65
GES-Net [80]	**75.87± 0.78**	82.37± 0.38
MN [13]	51.36±0.68	54.27±0.41
S2M2 [77]	69.00±0.41	82.14±0.21
MAML [20]	51.06±0.21	65.83±0.17
RN [16]	62.70±0.43	78.25±0.34
PCFGNet [85]	72.41 ±0.91	85.26 ± 0.66
PN [15]	58.62±0.41	80.63±0.39
DeepEMD [32]	63.76±0.74	78.25±0.43
DLA-MatchNet [26]	68.27±1.83	79.89±0.33
CovaMNet [29]	59.06±0.73	70.29±0.62
SCL-MLNet [74]	63.36±0.88	77.62±0.81
LLSR [51]	57.1	70.65
Baseline (Conv-4) [81]	71.42±0.33	**86.12±0.16**
SIB [83]	55.84±0.65	65.96±0.48
TSC [76]	70.99±0.74	82.18±0.32
CAN+T [84]	69.79±0.56	79.71±0.22
MMML (ours)	**77.76±0.69**	**88.40±0.54**

TABLE IV
CAs (%) OF MMML ON THE UCM DATASET

Method	Five-way	
	One-shot	Five-shot
MN [13]	34.22±0.77	53.12±0.20
S2M2 [77]	56.42±0.40	71.97±0.27
Meta-SGD [75]	51.78±1.05	65.47±0.65
DeepEMD [32]	58.47±0.76	70.42±0.58
TPN [78]	53.36±0.77	68.23± 0.52
RN [16]	48.60±0.73	62.34±0.66
PN [15]	52.36±0.21	68.97±0.72
DLA-MatchNet [26]	53.76±0.62	63.01±0.51
MAML [20]	49.03±0.28	60.89±0.27
SCL-MLNet [74]	51.37±0.79	68.09±0.92
LLSR [51]	39.47	57.4
TAE-Net [82]	**60.21± 0.72**	77.44±0.51
RS-MetaNet [53]	52.31±1.12	67.21±0.44
CovaMNet [29]	53.76±1.25	71.18±0.39
TSC [76]	55.11±0.68	69.20±0.64
SPNet [24]	57.64±0.73	73.52 ±0.51
MKN [54]	58.45 ±0.54	**77.92± 0.32**
MMML (ours)	**61.42±0.34**	**79.95±0.16**

and the most abundant scene categories. Hence, the experiment can well test the distinguishing ability of our method. We can observe that the methods (e.g., MN [13], RN [16], pn [15], Meta-SGD [75], and MAML [20]) developed for natural image recognition have poor performance when they are directly applied to RSSC, and this phenomenon may be caused by the complexity of remote-sensing images. What is more important is that our proposed method achieves optimal performance. Specifically, in the case of a five-way one-shot, the second best method is IDLN [79], and its CA is 75%. The CA of our method reaches 77.35%, which is 2.1% higher than IDLN. In the case of the five-way five-shot, among the previous methods, PCFGNet [85] achieved optimal performance, the CA of which is 85.07%. However, the method of our method is 88.19%, which is 3.12% higher than PCFGNet. In addition, the previous methods are directly implemented in Euclidean space, and only our method maps the features of Euclidean space to Riemannian manifold space. It is obvious that our method has more advantages.

Subsequently, we conducted experiments on the WHU dataset. The results are presented in Table III. As can be seen from the table, our method is still competitive. In the case of a five-way one-shot, MMML is the best method, the CA of which is 77.76%. The CA of the second best method (i.e., GES-Net [80]) is 75.87%. There is no doubt that the CA has been improved by 1.89% through our method. In the case of a five-way five-shot, the second best method is Baseline (Conv-4) [81], the CA of which is 86.12%. Compared with our method, the CA of which is 88.40%, the CA of GES-Net is lower than 2.28%.

Finally, to further demonstrate the advantages of our method, we also conducted experiments on the UCM dataset. Table IV shows the results of the comparative experiment. In two groups of experiments, our method is still the best. When the experimental setting is a five-way one-shot, through our method, the CA was improved by 1.21%, which increased from 60.21% to 61.42%. The second best method is TAE-Net [82]. When the experimental setting is five-way five-shot, MKN [54] is the second best method, the CA of which is 77.92%. The CA of MMML is 79.95%. Hence, our CA is improved by 2.03%.

From the perspective of datasets, our method achieved the most obvious improvement on the NWPU dataset. This is mainly because the number of samples in each category is sufficient, which is beneficial to the training of the model. From the perspective of experimental settings, the improvement of CA under a five-way five-shot is greater than that under a five-way one-shot. This is mainly because of the influence of the number of positive sample pairs and negative sample pairs. In the case of a five-way one-shot, each category of the image has only one positive sample pair and four negative sample pairs, which will affect the optimization of the distance. Through analysis, compared with other methods, our method achieves the best performance in the RSSC task. Our method has two main advantages. On the one hand, we metric the distance on two heterogeneous and complementary Riemannian manifold spaces, which enhances the representation ability of the features. On the other hand, we maximize the ratio of interclass divergence and intraclass divergence to optimize the distance, which can reduce the impact of large intraclass differences and high interclass similarity on scene recognition.

To further verify the effectiveness of our proposed method on each category, we calculate the confusion matrices on three test sets and visualize them. Fig. 3 shows the visualization results. We assume that i denotes the row numbers and j denotes the column numbers, the elements on the diagonal represent the proportion of correctly classified samples in each category. The other elements represent the proportion of the misidentified samples in each category (i.e., the samples in the ith class are identified as the samples in the jth class). From Fig. 3, we can observe that the CAs of all categories have exceeded 70% under the five-way five-shot setting. In the case of a five-way one-shot setting, only two categories of CAs are less than 60%, that is, beach and river in Fig. 3(c), the CAs of them are 56% and 52%, respectively. This is mainly because beach and river have many similar properties, both

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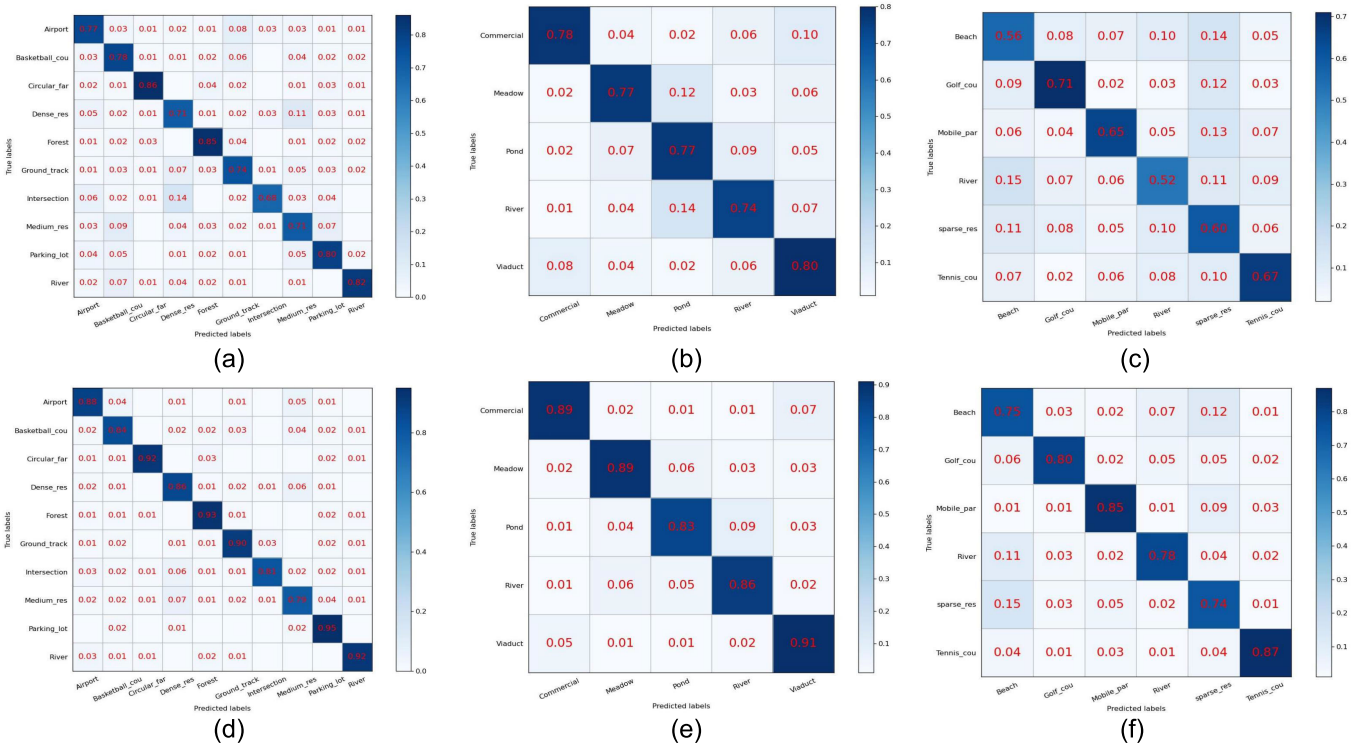


Fig. 3. Confusion matrices on the three datasets. (a)–(c) Confusion matrices on the NWPU, WHU, and UCM datasets in the setting of a five-way one-shot, respectively. (d)–(f) Confusion matrices on the NWPU, WHU, and UCM datasets in the setting of a five-way five-shot, respectively.

their images contain water bodies. In addition, there are bare lands near some rivers, which are easily identified as beaches.

In addition, to investigate the performance of our method on samples that are easily misclassified, we use different methods to perform comparative experiments on samples that are easily misclassified. The images are randomly selected from some difficult categories. We choose several classical methods (SPNet [24], pn [15], and MAML [20]) to predict the labels of the images. The results are shown in Fig. 4. We can observe that the palace is misclassified as a church under all methods. Because it is difficult to be distinguished. Even for humans, the spire building in the palace scene is easily mistaken for a church. SPNet recognized the intersection as the basketball court. For stadium, intersection, wetland, and medium residential, all of them can be identified correctly by MMML. Pn mispredicted the labels of stadium and wetland. MAML mispredicted the labels of intersection and medium residential. It is difficult to be identified correctly for some scenes, which is due to the interclass similarity. The results of comparative experiments indicate that our method is superior to SPNet, pn, and MAML.



Fig. 4. Comparison of classification results of different methods on samples that are easily misclassified.

D. Ablation Analysis

To demonstrate the effectiveness of all components in our proposed MMML, we implement comprehensive ablation learning. We use Grass and SPD to represent the Grassmannian manifold and SPD manifold module, respectively, and use D_L to represent the learnable distance metric module. In all ablation learning schemes, the feature embedding module remains unchanged. In the scheme where D_L is ignored, we directly use the distance defined in Sections III-A and III-B to calculate the similarity between the samples in query set and the prototypes. Specifically, we consider the following schemes.

1) *Grass:* We only consider the Grass component, ignoring SPD and D_L.

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TABLE V
CAs (%) OF THE ABLATION EXPERIMENTS ON THREE DATASETS

Settings	NWPU		WHU		UCM	
	One-shot	Five-shot	One-shot	Five-shot	One-shot	Five-shot
Grass	73.23±0.16	82.24±0.30	72.88±0.19	82.03±0.67	57.49±0.55	74.7±0.48
Grass+D_L	74.62±0.70	84.31±0.51	74.81±0.46	84±0.34	58.71±0.43	76.33±0.51
SPD	73.9±0.42	83.56±0.16	73.43±0.67	83.16±0.12	58.5±0.26	75.93±0.42
SPD+D_L	75.31±0.36	84.83±0.34	75.01±0.41	85.23±0.38	60.29±0.4	77.48±0.33
Grass+SPD	76.24±0.29	85.95±0.64	75.96±0.27	86.35±0.16	61.07±0.58	78.83±0.12
Grass+SPD+D_L(Ours)	**77.35±0.73**	**88.19±0.47**	**77.76±0.69**	**88.40±0.54**	**61.42±0.34**	**79.95±0.16**

TABLE VI
IMPACT OF WEIGHT FACTORS OF THE SPD MANIFOLD AND THE GRASSMANNIAN MANIFOLD ON CAs (%) IN THE CASE OF FIVE-WAY FIVE-SHOT ON THE THREE DATASETS

λ_1/λ_2	0.1/0.9	0.2/0.8	0.3/0.7	0.4/0.6	0.5/0.5	0.6/0.4	0.7/0.3	0.8/0.2	0.9/0.1
NWPU	84.44	85.63	86.59	87.35	87.94	**88.19**	87.84	87.56	87.45
WHU	85.07	85.99	86.87	87.52	88.21	**88.4**	87.92	87.54	87.24
UCM	76.3	76.68	77.43	78.32	79.07	**79.95**	79.36	78.76	78.6

2) *Grass + D_L:* We only add a D_L to the Grass component.
3) *SPD:* We only consider SPD.
4) *SPD + D_L:* We add a D_L to SPD.
5) *Grass + SPD:* We consider the combination of these two manifolds, but only use the fixed distance to measure the similarity.
6) *Grass + SPD + D_L:* that is, our proposed method, which takes into account all the components. The experimental results are shown in Table V.

It can be seen from Table V that the combination of Grass, SPD, and D_L achieves satisfactory performance. The CAs obtained on SPD are higher than those on Grass when only one manifold component is considered and the similarity is measured by a fixed distance. When the D_L component is added, the CAs of Grass and SPD are significantly improved. This is because D_L can optimize the distance according to the divergence of interclass image pairs and intraclass image pairs, which reduces the influence of the intraclass difference and the interclass similarity of remote-sensing images on scene classification. In addition, the CAs of Grass + SPD are higher than those of Grass, Grass + D_L, SPD, and SPD + D_L, which indicates that the combination of Grass and SPD can obtain a better effect than the single method, and two heterogeneous and complementary Riemannian manifold descriptors can enhance the feature's representation ability. The MMML we proposed is based on Grass + SPD, and the D_L component is added, then the performance is further improved. Under the setting of five-way five-shot, the CA of Grass + SPD + D_L on the NWPU dataset (its CA is 85.95%) is improved faster, which is 2.24% higher than that of Grass+SPD (its CA is 88.19%). Under the five-way five-shot setting, the CA of Grass + SPD + D_L on the WHU dataset (its CA is 75.96%) is improved faster, which is 1.8% higher than that of Grass + SPD (its CA is 77.76%). Through comprehensive analysis of the ablation experiment results, it is shown that the Grass, SPD, and D_L components proposed by us in MML all contribute to improving the CA of the remote-sensing scene, and their combination can produce the best results.

E. Hyperparameter Analysis

In this section, we will investigate the effect of hyperparameters on CAs. According to Section IV-B, we have three hyperparameters that need to be determined through repeated experiments, that is, m, λ_1, λ_2, and $\lambda_1 + \lambda_2 = 1$, where m denotes the dimension of the linear subspace. According to Section III, $0 \leq m \leq 196$. We set it as a series of discrete values to study its effect on CAs, that is, $m \in$ [20, 40, 60, 80, 100, 120, 140, 160, 180]. λ_1 and λ_2 represent the weight factors of the SPD manifold and Grassmann manifold, respectively. For λ_1 and λ_2, we set them as some pairwise weights to discuss their effects on CAs, that is, λ_1 and $\lambda_2 \in$ [0.1, 0.9], [0.2, 0.8], [0.3, 0.7], [0.4, 0.6], [0.5, 0.5], [0.6, 0.4], [0.7, 0.3], [0.8, 0.2], [0.9, 0.1]. Considering that it is difficult to optimize the three hyperparameters simultaneously, and λ_1 and λ_2 are interdependent, we first optimize m and then optimize λ_1 and λ_2. Specifically, we first assume that the two manifolds are of equal importance to CAs, that is, $\lambda_1 = \lambda_2 = 0.5$ and then determine m by performing repeated experiments. After obtaining the optimal m, we will further investigate the importance of λ_1 and λ_2 to CAs.

Fig. 5 shows the influence of the change of linear subspace dimension m on CAs. We can observe that with the increase of m, the CA curve shows a trend of increasing first and then decreasing slowly on the three datasets. The main reason for this phenomenon is that the lower dimension subspace provides less feature information, but with the increase of the subspace dimension, redundant information will also appear. Both of these two cases will affect the final classification results. For NWPU and UCM datasets, CAs reach the highest when $m = 120$. For the WHU dataset, when $m = 100$, CAs reach the highest. Hence, for the NWPU, WHU, and UCM datasets, the values of m are set to 120, 100, and 120, respectively.

Table VI shows the impact of the weight factors of the two kinds of manifold features on the experimental results. We can see that the CAs show a similar change trend on the three datasets. When $\lambda_1 < 0.6$, CAs increase with the increase of λ_1. When $\lambda_1 = 0.6$, CAs reach the highest. When $\lambda_1 > 0.6$,

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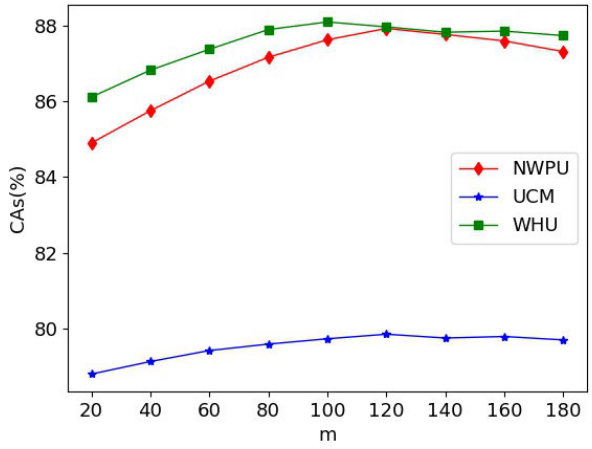


Fig. 5. Effect of different m on CAs in the case of a five-way five-shot on the three datasets.

CAs begin to decrease slowly. Hence, when $\lambda_1 = 0.6$ and $\lambda_2 = 0.4$, our model can achieve the best performance.

V. Conclusion

In this article, we propose an MMML method for RSSC with FSL. The motivation of our proposed method is to reduce the interference of interclass similarity and intraclass difference on RSSC, then the CA can be improved. Specifically, first of all, to further enhance the representation ability of the features, we embed the feature map of the CNN into two heterogeneous and complementary Riemannian manifold geometric structures, that is, Grassmann manifold and SPD manifold, and construct descriptors on manifold space to form the Grass and SPD components. Then, we propose an optimization strategy based on positive sample pairs and negative sample pairs when designing the distance measurement schemes. We use the embedding features of the samples in the query set and the embedding features of the same category in the support set to construct intraclass pairs and calculate the intraclass divergence. Subsequently, the embedding features of the samples in the query set and other embedding features which do not belong to this category in the support set are used to construct interclass pairs to calculate interclass divergence. The ratio of interclass divergence and intraclass divergence is used as an optimization function for calculating distance, and we should maximize it, thereby reducing the impact of large intraclass differences and high interclass similarity on scene recognition. In addition, we also propose a Riemann kernel function strategy to fuse two different kinds of manifold descriptors. Additional ablation experiments further demonstrate that the Grass, SPD, and DL components can help improve the accuracy of RSSC, and combining them simultaneously can achieve the best results.

Although our method has achieved good performance in few-shot RSSC, it also has a limitation. Our method can only utilize labeled samples for training. As is known to all, the cost of obtaining labeled samples in remote sensing is very high. In future research, it is worth considering combining our method with unsupervised learning and self-supervised learning techniques to fully exploit the information of unlabeled samples and further reduce the dependence on labeled samples in the training process of RSSC.

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